

Phase IV Deliverables for Objectives H, J and K
(Energy Component)

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1. Introduction

The objectives being addressed from the MoU are:

- **H. Calculation of water productivity**
- **J: Design of extension program to improve pump selection**
- **K: Village level irrigation and energy infrastructure documentation**

The deliverables of Phase IV are:

- H.3 Estimates of water and energy off-season consumption
- J.1 Report on current practices of water infrastructure, pump set and pipeline selection.
- J.2 Analysis of operational efficiencies of existing pump sets on the field, and guidelines for pump and pipeline selection.
- K.1 Irrigation/ energy infrastructure analysis along with water transfers structures

Chapter 2 deals with pump selection study [J1, J2] and chapters 3 and 4 include irrigation infrastructure analysis, water transfers and energy consumption [H3 and K1].

2. Pump Selection Extension Program

The poor operational efficiency of agricultural pumps is a well-documented phenomenon, attributed in a very broad sense to low tariffs for agriculture. There have been numerous high efficiency pump dissemination programs over the last 50 years to address this gap. However, it is not clear what effect these programs have had at the end of the program, partly due to a lack of documentation or its availability in the public domain, and due to an absence of holistic or extended evaluation (Prayas (Energy Group), et al., 2018). Most of them have been pilot projects conducted in pockets, and is unlikely that the lessons have been incorporated into any lasting practices or policies (Khobragade, 2018) (Prayas (Energy Group), et al., 2018). In addition, the energized irrigation sector in Maharashtra is very dynamic, due to changing cropping and irrigation practices, as well as the steady growth in the number of pumps being used.

Selecting an appropriate pump is the most effective part of energy efficiency and electricity distribution infrastructure cost reduction, in the case of pumps. It is also very important for proper irrigation management. Hence, this project has been undertaken with a view to analyse Irrigation Pumping System (IPS) selection and usage in Vidarbha and Marathwada, and thereby develop an extension program for the same, as well as give higher level feedback for policy changes.

This report includes the insights from primary interactions with various stakeholders, results from the on-field efficiency measurements with different irrigation systems, comparison of existing pump sets with the actual practical need, case studies of pump selection and lab exercise to understand variation in pump performance wrt changes in voltage and loads. The design of the extension program for improving IPS selection will be delivered in Phase V.

Objectives of this project:

1. To design an extension program for appropriate pump selection which will accomplish the following:
 - Reduce energy consumption
 - Reduce the load on distribution system due to pump sets being downsized
 - Give farmers confidence in the pump selection process
2. Identify improvements in labelling on pumps to make selection easier

3. Identify avenues / institutions to codify these practices of pump selection, efficiency measurements, labelling

2.1. Executive Summary of the outcomes:

- 48 farmers, and 9 vendors, from the districts of Aurangabad, Buldhana, Hingoli, Washim, Osmanabad and Yavatmal districts were surveyed to find out how farmers select pumps.
- Quality of electricity supply is one of the most important factors for all farmers:
 - Pumps should have reasonable performance at low voltages.
 - Pumps should have high enough flow rates so that irrigation can be completed quickly due to the lack of reliability of the quality and duration of electricity supply.
- Cost matters, and the local pumps are cheaper. However, there is a greater infiltration of ISI mark pumps with programs like PoCRA, and also because these manufacturers have adapted products to higher performance at low voltages.
- Farmers follow the advice of the vendors, plumbers, and electricians for selection. Vendors stated that they ask the farmers details about field distance from well, depth of well, pipe diameter, and recommend pumps based on broad requirements.
- Recently, Texmo Company conducted a training session for shop owners and pump installers in Buldhana. Companies like Texmo and CRI have developed pump selector programs (software).
- An analysis of 31 farmers who had recently purchased pumps indicated that 29% farmers had oversized pumps, hence the load could reduce by 2 kW per farmer, leading to an overall load reduction of 13%.
- 29 out of 31 pumps were not selected for the appropriate duty point and a detailed analysis shows a potential saving of 15% energy.
- PoCRA's subsidy is such that the farmers need to spend Rs. 4000 – 6000 (not a big difference) whether it is 3 HP or 5 HP, and hence sometimes tilts them towards 5 HP.

- It was observed that many farmers can satisfy their requirement with a capacity less than 5 HP, but they select a 5 HP pump or higher due to the fact that MSEDCL has the same tariff for 3 and 5 HP pump sets. The current tariff structure has been set according to (0 to 5 HP, 5 to 7.5 HP, and > 7.5 HP). Taking into account the 15% reduction in energy by using 3 HP capacity pumps instead of 5 HP, a one more category of 0 to 3 HP pump in addition to the current tariff structure can have a significant impact on the pump selection criteria.
- The standards for Ag pumps specify very basic information for labelling. Some information is not given to farmers unless they ask for it. In addition, this information format is complex, and should be presented in a farmer-friendly manner, in the local language, if farmers are to be more aware of proper pump selection.

2.2. A brief description of pump characteristics

A pump has varying efficiencies and flow rates at different heads. An appropriate selection means, selecting a pump such that the operating head, or the head in the field (depth of water + frictional head + pressure required for sprinkler/ drip) matches with the duty point of the pump, or the head where the efficiency is maximum (Best Efficiency Point). Also, the HP should be such that the pump can supply the required flow rates at the operational head, and yet the HP should not be more than required.

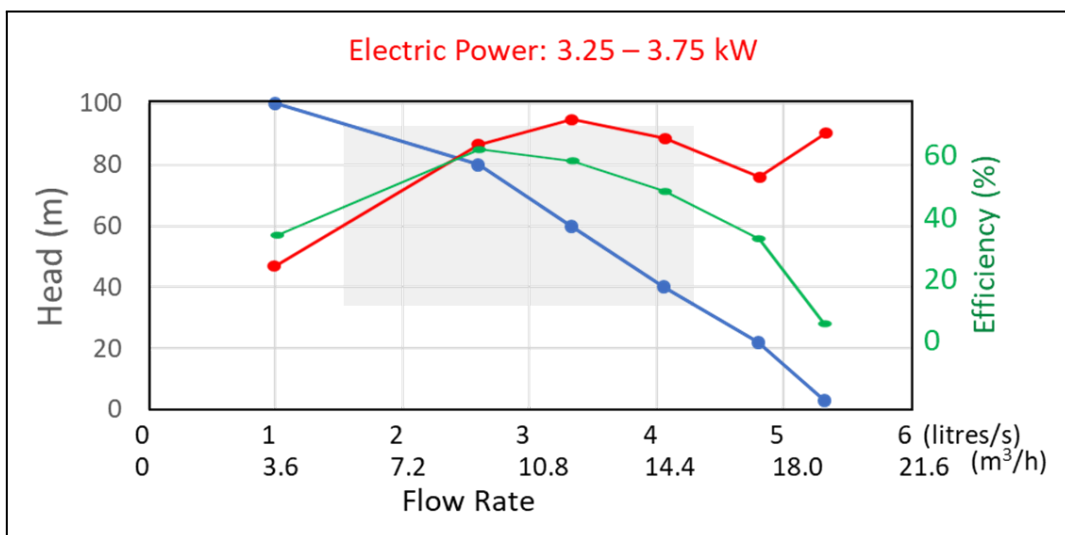


Figure 1: Characteristics of a Texmo borewell pump measured in the lab. The grey box indicates head and flow rates available in the company catalogue. This is a 5 star rated pump.

Figure 1 shows a sample pump characteristic. The Best Efficiency Point of this pump is at 80m, and the range as specified by the manufacturer (over which standards are met) is indicated by the grey box. If the pump was only required to run at a single point or a small range, selecting a pump would be straightforward. However, it is sometimes possible that the range over which the pump is required to run is much larger.

Farmers use a pump for furrow and the same pump is used for drip/ sprinkler where the head increases by 25 – 30m (this is in addition to variation in the level of the water), but the flow rate is much lower. This could create a problem in selection because often the characteristics provided by the manufacturer do not indicate the lower side of flow rates. Hence the farmer may purchase a larger than necessary pump considering a very high head.

PERFORMANCE CHART																			
TEXMO "TSM-Z SERIES" - THREE PHASE OPENWELL SUBMERSIBLE MONOBLOCKS (STAR)																			
Approximate performance values of TSM-Z series - Star at 415V (-15% to +6%), 2900 rpm, 50 Hz AC power supply																			
Model Name	Connection	Motor Rating		Pipe Size (mm)		TOTAL HEAD VALUES													
						Metres	6	8	10	12	14	16	18	20	22	24	26	28	30
		kW	HP	Suc.	Del.	Feet	20	26	33	39	46	52	59	66	72	79	85	92	98
TSM 30 FZ 	DOL	2.2	3	65	50	↑							9.0	8.2	7.2	6.0	4.4		
TSM 1 Z 	DOL	2.2	3	65	65					12.4	10.9	9.3	7.8	6.2	4.1				
TSM 1 FZ 	DOL	2.2	3	65	65					13.7	12.4	11.0	9.3	7.6	5.1				
TSM 12 Z 	DOL	2.2	3	75	65			18.7	16.7	14.3	11.7	8.1							

Figure 2: Series of Texmo 3 HP submersible open-well pumps

For example in Figure 2 the characteristics stop at 4.4 lps at a 26m head, whereas drip/ sprinkler have been observed operating at 2.2 lps. But it will not be clear from this catalogue that this pump could operate a drip/ sprinkler for a certain configuration, and a farmer may select a 5 HP pump even when a 3 HP pump may fulfil the need.

2.3. Investigation of existing pump selection practices of farmers

An open ended survey was conducted to understand how farmers decide the IPS which satisfies their requirements. After identifying several important factors that farmers assess, the questions were then narrow downed to find out the most influential factors in IPS selection. It is interesting to see priorities changing across different regions. The distribution and criteria of IPS selection of 48 farmers surveyed from Aurangabad, Buldhana, Hingoli, Washim, Osmanabad and Yavatmal districts are summarized in Table 1 and Table 2.

Table 1: Villages from which data was collected

Villages	Taluka	District	No. of farmers surveyed
Garaj	Vaijapur	Aurangabad	6
Dhorapgaon	Khamgaon	Buldhana	5
Shirad Shahapur	Aundha Nagnath	Hingoli	5
Umbarda, Yevta, Manbha	Karanja Lad	Washim	18
Junoni, Walgud	Osmanabad	Osmanabad	5
Kondhai, Yehala	Pusad	Yavatmal	9

Table 2: Criteria of pump set selection by farmers

Criteria	Rationale
1. Able to operate at low voltages (as low as 140 V)	- Poor quality of electricity supply during peak season (November-January) due to increased competition and very low load diversity factor - Delivers high or desirable flow rate at low voltages - Low initial investment required
2. Gives high flow rate (specifically for furrow irrigation, 5 to 8 lps)	- High uncertainty of electricity supply - Fear of crops facing shortage of water when needed resulting into low yields in terms of quality and quantity - Need to cover the entire field within a given period of time of power supply. Again increasing the competition for water and energy resources.
3. Method of irrigation (specifically for sprinklers)	- If a sprinkler system is present, farmers want to use more number of sprinkler nozzles at a time to cover more area in 1 shift. (number of nozzles vary from 9 to 16 depending on nozzle type and farm size/ dimensions) *This criteria is ignored in case of farmer using furrow irrigation

4. Able to deliver water at long distances (1 to 3 km)	- Expect high flow rate as farmers need to cover entire field within a given period of time of power supply - This is either used for direct irrigation or for water transfers
5. Area to be irrigated, type of water source (well: depth of well) and cropping pattern	- 5 HP and more if the land holding is more, distance between water source and field is more (typically >1000 ft.), and crops are diversified throughout the year
6. Investments	- Constant uncertainty of water and energy availability lead to selection of local pump sets as they are cheaper. - Some farmers see pump sets as a short time investment as they need it only for 3-4 months. - On the other hand, when selecting a pump set through a subsidy (like PoCRA), a general behaviour was seen to select a higher capacity pump. This is because the actual investment amount remains almost the same (Rs. 4000 to 6000/-) with 3 HP and 5 HP capacity pump. - There are farmers who keep extra pumps as a backup or use them at the same time to irrigate more area within a given time.

- 70 % of the farmers ranked the first three factors as most important and what they look for in a given pump set. Low voltage operation is the topmost criteria amongst all, common for all regions.
- Selecting pump set having 5 HP and more capacity was found to be most prominent in farmers who lift water from Dam or a percolation tank in Yavatmal and Washim for moving water over longer distances.
- About 50% of the farmers also consider the method of irrigation (specifically for sprinkler) they are already using or are planning to put in place. This is the case particularly from fellow farmers experience and when asked by the pump vendor.
- There is a typical correlation farmers follow between pump HP and pipeline diameter from their own experience and in consultation with pump vendors. If the pipeline is already in place, then the pump capacity is selected according to the pipe diameter. (5 HP for 2-2.5" pipe diameter, 7.5 HP for more than 2.5" pipe diameter). Few of the farmers having 2" pipe diameter pipeline and 5 HP pump set also reported that, they have to attach a bypass valve to compensate for the pressure. This is to avoid loosening and bursting of pipelines.
- While cost is always a deciding factor during pump selection, the trend is now seen shifting towards buying an ISI marked pump sets. Some of the reasons are they have

improved performance at low voltages, the mandate given by various government programs such as PoCRA and a word of mouth about a particular pump manufacturer.

- None of the farmers from the surveyed sample goes into technical details about pump duty point, operational characteristics and what standard accessories can be used to improve the performance or reduce pump burnouts.

2.4. Investigation of pump selection methodology of pump vendors

Role of a pump set distributor is to calculate the required head and flow rate based on the information provided by farmers. This information contains details about type of water source, depth, area to be irrigated, pipeline details and irrigation method used. About 9 pump distributors were visited to gauge their methodology in pump selection.

Based on the information provided by farmers, if open well depth is within 50 ft. and pipeline length is within 2000 ft. with 2.5” diameter, then the total dynamic head is calculated first (5 to 7 lps flow rate is assumed for calculations). From the calculated value, a 5 HP capacity model having 30 to 36m head as rated value is suggested by the pump distributors. If the pipeline length is greater than 2000 ft. then pump sets rated head value more than 30m are suggested.

Hence, a regional generalization in the selection process is there as pump distributors have an overall idea of water sources, well depths, soil types, energy availability and quality, and cropping pattern.

Pump selector programs developed and owned by Texmo and CRI companies are being used by one of the pump distributors in Buldhana. Recently, Texmo Company conducted a training session for shop owners and pump installers in Buldhana. The objective was to create awareness about pump calculations.

Table 3: A general criteria followed by one of the pump set distributors

Length of the pipeline	Pump capacity and Head		
	3 HP	5 HP	7.5 HP
< 2000 ft.	26 m	30 to 36 m	-
2000 to 3000 ft.	44 m	48 m	48 to 56 m
> 3000 ft.	-	48 m	48 to 56 m

**This approach is followed by a distributor selling Lubi company's pump sets in Washim.*

2.4.1. Summary of Investigations

- Role played by the local plumber and/ or electrician and the pump vendor in the entire pump set selection process is significant in deciding the particular model of the same HP, the pump manufacturer and other accessories.
- During on field measurements in September-October 2020, actual power ratings are found to be higher than the sanctioned load as told by the farmers (e.g. 4 to 5 kW for 3 HP pump). This might be because of installation of higher capacity pumps (oversized) or high energy consumption of locally made non-ISI market pump sets. This was different from what we found during the initial visit in February 2020, that the input power for a 3 HP pump varied from 2.3-3.8 kW and that for a 5 HP varied from 3.8-5.7 kW. Hence we can say that, there are farmers using pumps of the stated rating as well as oversized pumps as compared to sanctioned load.
- The head range is wide for many farmers which is due to diversified crops with different irrigation methods. The pump capacities were suitable for furrow irrigation of most of the crops.
- Variation in the water column height (mbgl) seasonally leads to the selection of pumps that can operate at a higher head or oversizing. But the operating efficiency was found to be less than 10% when operated at low heads and in case of surface irrigation (during Kharif and at the start of Rabi season).
- Poor quality of electricity supply during peak season (November-January) including frequent tripping, DT failures, forces farmers to go for pumps that are able to run at low voltages and giving desired flow rate over the higher static and dynamic head. This in turn leads to poor efficiencies, poor performance of irrigation systems, higher investments and wastage of time, energy and money.
- Farmers also keep more than 1 pump set as a backup option in case of pump burnouts. 40 out of 48 surveyed farmers (~85%) have more than 1 pump set.
- Additional head of drip or sprinkler system is taken into account when suggesting a pump. However, it was also found out that if a farmer wants to install a drip or sprinkler system on the existing pump set, then the pump is considered to match up with the additional head and deliver required output. This raises a question about oversizing of existing pump sets.

- Energy aspect is generally disregarded by the farmers while pump selection due to lack of awareness about energy efficiency, technological attributes and low electricity tariffs.

There is no straight answer to farmers' agriculture pump set selection criteria, however a general inclination is towards low voltage compatibility and high flow rate expectation at a given head. Other two parameters are previous own experience and knowledge gained from the fellow farmers for certain pump HP.

2.5. Operational efficiencies of pump sets on field

Now that we have an overall idea of common pump selection practices by farmers, the second important task is to find out the operating characteristics (variation of flow rate with depth of water) and efficiencies of the pump sets installed by the farmers.

One of the important determinants of the efficiency of a pump depends on the best efficiency point, or the pressure head it is rated for, and the pressure head it actually operates at in the field. Other determinants are operating voltage, age of the pump, etc. The head will depend on the depth of water which varies with season, as well as with time during the irrigation cycle. Operating head also depends on the piping infrastructure, drip or sprinkler system will also add an additional head.

A sample set of 20 farmers was selected representing different soil types, crops, irrigation methods, water column heights, and distance from source to field, etc. Instantaneous measurements were taken for 8 farmers and for the rest 12 farmers who were irrigating their fields, 3 to 4 measurements were taken throughout a day to track variation in the efficiency wrt head.

In order to understand the complete operating characteristics and efficiencies of existing IPS, it is important to analyse the case when pumps are operating at low voltages and high static head in the peak season. Follow up measurements on the same pump sets in Washim and a similar study in Aurangabad and Buldhana are planned in Rabi and summer season.

The head also depends on the depth of water (or water column height i.e. static head) which varies with season, as well as the time in the irrigation cycle. This is because, when the

farmer begins to irrigate the water level is high, but the level drops at the end of the day due to the fact that the rate of pumping out is more than the yield of well (inflow of water).

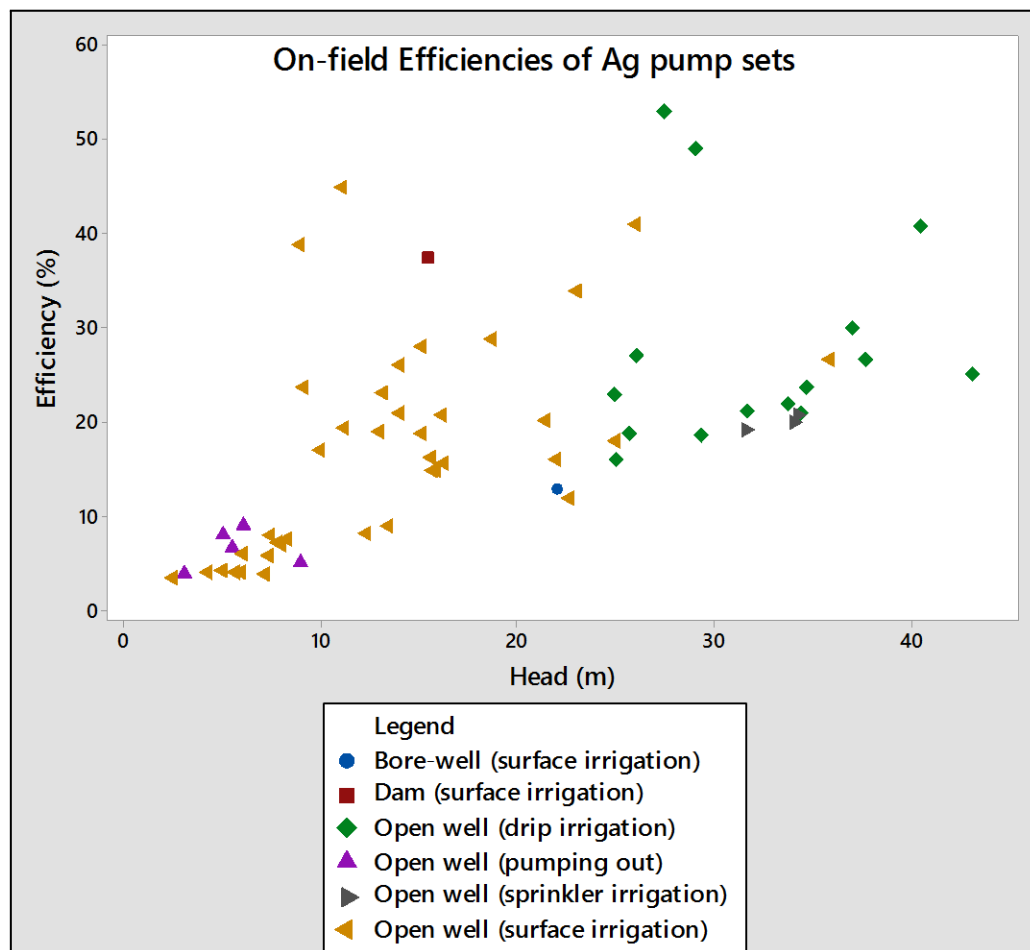


Figure 3: On-field Efficiencies of Ag pump sets (measurements taken on the field in the month of Feb 2020, Sept/Oct2020 in Washim cluster and Dec 2020 in Aurangabad cluster)

Figure 3 indicates variation in pump set efficiencies wrt total dynamic head for different irrigation systems (ISI as well as non-ISI rated pump sets). There are 63 measurements taken on-field in Washim and Aurangabad clusters. Average efficiency is found out to be 19% with a standard deviation of 11.8.

For heads less than 10 meters, pump efficiencies are found out to be less than 10%. One of the reasons for this can be very low head value and hence low hydraulic output for the same electrical input. Less than 30% efficiencies were calculated for surface irrigation methods except for few farmers who have high frictional heads (due to large distance between source

and field). While in case of pressurized operation (connected to drip and sprinkler system), the operational efficiencies were found to be in the range of 20 to 55%.

An example of efficiency variation wrt change in static head is shown in Figure 4 (Farmer 2) and when drip or sprinkler system is connected to a same pump which was initially operating for surface irrigation (for both Farmer 1 and Farmer 2).

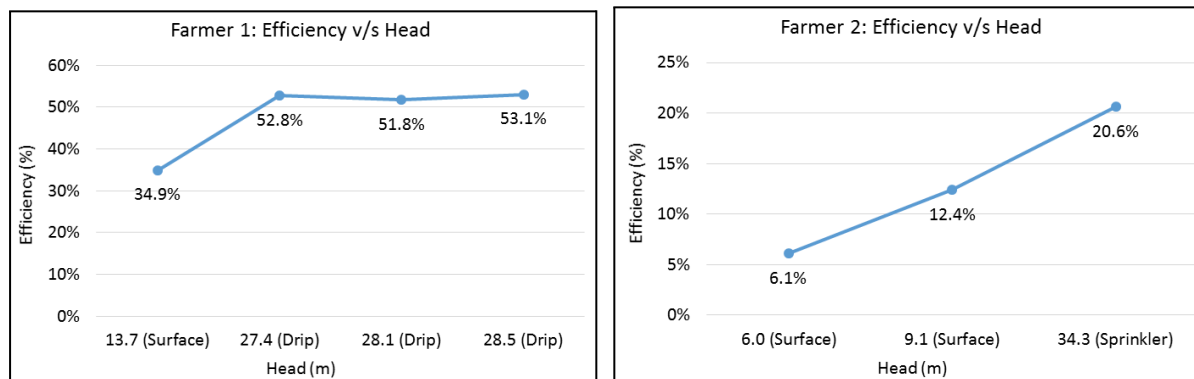


Figure 4: Efficiency variation when drip and sprinkler system are connected to the same pump (Farmer 1: ISI rated pump. Farmer 2: Non-ISI rated pump)

For farmer 2, the efficiency has almost doubled in this case. The reason for this can be that the existing pump set has been installed to operate efficiently at higher heads or has head range inclining towards higher head values.

2.5.1. Comparison of operational efficiencies with pump characteristics

On-field measured efficiencies of ISI rated pump sets that are subsidized through PoCRA and otherwise are tabulated in Table 4. (In Aurangabad and Washim village clusters)

Table 4: On-field efficiencies of ISI marked pump sets

Farmer	HP rating	Total dynamic head (m)	Flow rate (m ³ / hr)	Electrical input measured (kW)	Mechanical output (kW)	Efficiency
*Farmer 1	3	28	12.6	1.8	0.96	52%
		14	18.6	2	0.71	35%
Farmer 2	7.5	40	20	5.3	2.15	41%
		38	14	6.3	1.43	27%
Farmer 3	5	39	16.1	5.7	1.71	29%
Farmer 4	5	36	15	5.6	1.48	27%
Farmer 5	5	5	23	5.0	0.34	7%
		34	12	5.0	1.11	22%
Farmer 6	7.5	21	19	5.6	1.12	20%
Farmer 7	5	29	12	5.2	0.98	19%
Farmer 8	5	16	19	5.0	0.82	16%
Farmer 9	5	16	19	5.4	0.80	15%
Farmer 10	5	7	16	5.4	0.31	6%
Farmer 11	5	5	14	4.4	0.19	4%

All measurements were recorded on farmers having dug well as a water source because static head could not be observed for bore wells. All measurements are done on open well submersible pumps.

*ISI market Mono-block pump set, not subsidized through PoCRA.

The average efficiency is found to be 23% and a standard deviation of 13%, with more than 35% efficiency measured in case of only 2 pump sets. The efficiencies are very low considering the ISI rated pump sets and even if the operating head is within the rated head range (for efficiency values of 19 to 29%). For other cases, one of the reasons could be operating a pump meant for high head application at lower head values. If compared to standards, the efficiencies are found out to be much lower than defined (35 to 58%) (BIS, 2018).

2.5.2. Insights from Garaj, Aurangabad field visit

- 65 pump sets are subsidized through PoCRA in the year 2019-20.
- It was observed that more than 50% of the beneficiaries (farmers) are not using the pumps purchased through PoCRA (ISI rated) and have kept these pump sets as a back-up option.
- Almost all of the farmers applied for a higher capacity (5 HP) pump irrespective of the actual requirement. There are 3 major reasons for this, MSEDCLs tariff structure (same tariff for 0 to 5 HP load), almost similar actual investments for 3 and 5 HP pump sets, and lack of awareness about pump selection practices.
- For some of the pump sets subsidized through PoCRA (applicable only for 3 and 5 HP pump sets), 7.5 HP rating is mentioned on the pump test report, but the bill is given for 5 HP pump set. The pump characteristics as received from the pump vendors and Agriculture Assistant barely match with the actual measurements.
- Also similar pump test reports and characteristics are maintained/ submitted for different pump sets. The maintenance of correct pump characteristics and test reports is a serious issue, and due to this limitation comparison of measured efficiencies with pump characteristics was not studied.

2.6. Case Studies on Pump Selection

In this section, we tried to find out the reasons for efficiency variation and understand the actual requirements of pump characteristics for a certain field configuration. It is necessary to find out the characteristics of the existing pump farmers are using, how they are using it, what is the energy consumption, costs associated with it, and if the farmers have selected an appropriate pump set.

Table 5 gives a general overview of this exercise with 31 farmers from different agro-climatic zones. The farmers who have recently purchased a pump through PoCRA and otherwise were selected randomly. After understanding the cropping pattern, irrigation practices and field configurations, verification is done if the selected pumps are appropriate or oversized (of higher capacity than required) for each of these farmers. If a pump is oversized, a best suited pump is suggested and an estimation of energy and cost savings is done.

From the exercise, 29% pumps (9 out of 31 samples) were found to be oversized. This can lead to reduction of at least 13% load (~2 kW per pump). 29 out of 31 pumps were not appropriately selected, and hence would operate inefficiently. Appropriate selection can lead to a 15% reduction in energy usage.

Table 5: General details of farmers included in the pump selection exercise

	Washim (Umbarda, Yevta, Manbha, Pikheda)	Aurangabad (Garaj)	Osmanabad (Junoni, Walgud)	Buldhana (Dhorapgaon)
No. of farmers in exercise	15	6	5	5
Cropping pattern	Soybean, Cotton, Legumes, Wheat, Gram, Orange, Lemon, vegetables, Sugarcane	Cotton, Maize, Ginger, Wheat, Onion, Gram, Vegetables, Sweet Orange	Soybean, Wheat, Sorghum, Onion, Gram, Vegetables	Cotton, Legumes, Soybean, Wheat, Onion, Gram, Pulses, Sugarcane, Lemon, Vegetables
Water sources	Open well, borewell, irrigation tank	Open well, bore well, farm pond	Open well, borewell, irrigation tank	Open well, bore well, Nala
Well depths	30 to 60 ft.	50 to 70 ft.	30 to 50 ft.	50 to 75 ft.
Irrigation practice	Mixed combination of all 3 methods	All 3 types. More farmers using Drip system	All 3 types. More farmers using Surface method.	All 3 types. More farmers using Sprinkler system.

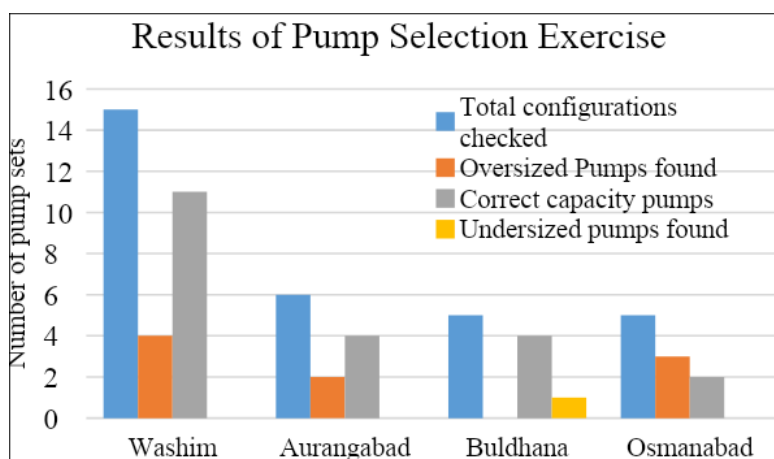


Figure 5: Results of pump selection exercise

As indicated in Figure 5, 9 oversized pump sets were found out from a sample of 31 farmers. These include ISI rated (subsidized through PoCRA) and non-ISI rated pump sets as well. The oversized pump results into more energy consumption than actually needed and thereby increasing the costs associated with it. Selection of oversized pumps also increases the initial investment for farmers. And if these pumps are subsidized through PoCRA, unnecessarily more subsidy for 5 HP pump set (75% of the total bill amount/ claimed amount, maximum subsidy allowed Rs. 15,000/-) gets sanctioned to a farmer who practically needs a 3 HP pump. (Rs. 5000 to 6000/- difference between the prices of 3 and 5 HP pump sets.)

On the other hand, considering a scenario where there are 20 farmers each sanctioned with 5 HP capacity pumps on 1 Distribution Transformer (DT) of 100 kVA capacity. If 5 of these farmers have oversized pumps than required or sanctioned, it will increase the load on that DT and eventually resulting in overloading problems and tripping. Therefore, this increases challenges on the MSEDCL side too.

Table 6: Example of energy and cost savings if 5 HP (oversized) is replaced with 3 HP pump

	5 HP pump	3 HP pump
Initial investment for a farmer (Rs.)	16,500.00	10,000.00
*Energy consumed (kWh/ year)	5,422	3,253
*Energy saved (kWh/ year)		2,169
*Cost saved for farmer (Rs./ year)		6,500.00
*Cost of energy savings (Rs./ year)		8,676.00
Subsidy cost saved (Rs.)		5,000.00 to 6,000.00
Cost saved for MSEDCL (Rs.)	Infrastructure cost (average Rs. 1 lakh) goes down and more connection can be managed	

*Energy consumption and cost savings are calculated for 1 acre of land. Detailed calculations are given in the Appendix II (Table 11 and Table 12).

2.6.1. Case Studies:

A detailed exercise was carried out for all 31 farmers checking the appropriateness of existing pump sets and selecting a best suited pump for given configurations. This is to build on the understanding from interviews about case to case as well as regional variation in selection of pumps and to make a comprehensive extension program. This exercise will also help in building some thumb rules for creating pump selection guidelines as there are a number of factors to be considered. 6 such case studies are mentioned in this section from a sample of 31 farmers.

Case I: Farmer 1: Umbarda bazar, Washim

Parameters	Details		
Soil type	Clayey loam		
Cropping details	Crop	Acres	Irrigation Method
	Cotton	2	Surface + Sprinkler
	Tur	4 and 1.5	Surface + Drip
	Wheat	3	Surface
	Gram	2	Sprinkler
	Guava	1.5	Drip
Water source details	Type:	Open well	
	Depth:	40 ft.	
	Diameter:	20 ft.	
	Water column change:	4 to 25 ft.	
Pipeline details	Material:	PVC + HDPE	
	Diameter:	2" and 2.5"	
	Length:	200 to 900 ft.	
Irrigation infrastructure details	Sprinkler:	9 nozzles	
	Drip:	1.5 X 4 ft. spacing	
Existing pump set details	Capacity:	5 HP	
	Model:	CRI: CSM 46 (ISI rated)	
	Head range:	46 to 29 m	
	Flow rate range:	8 to 14 lps	
	Overall efficiency:	40%	
Actual characteristics required for given configurations	Head range:	35 to 15 m	
	Flow rate range:	2 to 6 lps	
	Pump Capacity:	3 HP	
Suggested model	Silver: M 101	3 HP	
	Head range:	36 to 12 m	
	Flow rate range:	2 to 7 lps	

Based on the field configurations, the operating head range is 35 to 15 m with flow rate requirement of 2 to 6 lps. But the existing pump set has a best operating head range of 46 to 29 m and 8 to 14 lps, which is very high compared to the actual requirement of the farmer. This is the typical case of selecting a pump with higher capacity (an oversized pump). As found out from the in-depth interview with the farmer, two major reasons for selecting the existing pump are the unreliable supply of electricity, and area under cultivation is more with a mix of irrigation methods. The pump vendors also support this fact of recommending a higher head pump keeping in mind the unreliable electricity supply, cropping pattern and water source details.

The on-field measurements in the month of February 2020, a flow rate of 12 lps was measured for surface irrigation method at 29 m head with < 30% efficiency. Farmer also said that the flow rate seems to be very high as it breaks the bunds of furrows or at times.

A 3 HP pump set as suggested will be an appropriate choice in this case.

Case II: Farmer 2: Yevta, Washim

Parameters	Details		
Soil type	Clayey loam		
Cropping details	Crop	Acres	Irrigation Method
	Cotton	4	Surface
	Tur	2	Surface
	Wheat	4	Surface
	Water Melon	1	Drip
Water source details	Type:	Open well	
	Depth:	60 ft.	
	Diameter:	16 ft.	
	Water column change:	10 to 40 ft.	
Pipeline details	Material:	HDPE	
	Diameter:	2.5" and 3"	
	Length:	3000 to 4000 ft.	
Irrigation infrastructure details	Drip:	1.5 ft. X 4 ft. spacing	
Existing pump set details	Capacity:	5 HP CRI: CV 1 HD (2 stage)	
	Head range:	41 to 24	
	Flow rate range:	8 to 13 lps	
	Overall efficiency:	50%	
Actual characteristics required for given configurations	Head range:	41 to 33 m	
	Flow rate range:	2 to 6 lps	
	Pump Capacity:	5 HP	

This pump set is correctly sized with head range and pump capacity both fitting to the actual requirements. The required head value is on a higher side majorly due to frictional head as the water is getting transferred over a distance of 4,000 ft. (~1.2 km) from open well to farm.

However, from the on-field measurements in September-October 2020, it was found that the flow rate was 4 lps at 39 m head with an overall efficiency of 32%. This might be because the flow rate measured (4 lps) was beyond the range of pump characteristics (8 to 13 lps). Hence, low efficiency was obtained even if the operating head is within the recommended range.

Follow-up flow rate and efficiency measurements will be taken for this farmer in different seasons (Rabi and summer) to check the on ground performance.

Case III: Farmer 3, Garaj, Aurangabad

Parameters	Details		
Soil type	Clayey		
Cropping details	Crop	Acres	Irrigation Method
	Cotton	6	Surface
	Maize	2	Surface
	Ginger	2	Drip
	Wheat	1	Surface
	Gram	2	Surface
Water source details	Type:	Open well	
	Depth:	60 ft.	
	Diameter:	20 ft.	
	Water column change:	5 to 30 ft.	
Pipeline details	Material:	PVC + HDPE	
	Diameter:	2"	
	Length:	200 to 300 ft.	
Irrigation infrastructure details	Drip:	1 X 5 ft. spacing	
Existing pump set details	Capacity:	5 HP	
	Model:	HO-82 (ISI rated)	
	Head range:	37 to 24 m	
	Flow rate range:	8.8 to 14 lps	
	Overall efficiency:	41.5 to 45.8%	
Actual characteristics required for given configurations	Head range:	40 to 16 m	
	Flow rate range:	3 to 6 lps	
	Pump Capacity:	5 HP	
		3 HP (if pipe dia increased)	
Suggested model	Silver: M 121	5 HP	
	Head range:	44 to 16 m	
	Flow rate range:	4 to 9 lps	

In this case, for the given configurations, the selected pump set matches in terms of pump capacity (5 HP), but the operating head range is very small (37 to 24 m). As the farmer is irrigating most of his crops with surface irrigation method, the total dynamic head in this case will vary between 16 to 20 m, again out of the existing pumps characteristics. Hence, a pump set with greater head range as suggested is suitable for this farmer.

Apart from this, the maximum head is when a drip system is connected to the pump. Existing pipe diameter is 2", with a frictional head of 5.5 m and additional 25 m head due to the drip system making it total 40 m head (static + frictional + drip). Flow rate required to cover the field using drip irrigation is 3.6 lps. This needs more electrical input (>3 kW) which is beyond the capacity of 3 HP pump, and hence a 5 HP capacity pump is suggested for this farmer.

However, if the pipe diameter is increased to 2.5" only in case of the drip system, the frictional head gets reduced to 1.8 m and a total dynamic head becomes 36 m with electrical input now required close to 3 kW at 40% overall efficiency. This will save the energy consumption and hence the cost associated with it. This will also reduce the initial investment of the farmer (Rs. 6000 to 7000/- cost difference between 3 HP and 5 HP pump sets).

Case IV: Farmer 4, Dhorapgaon, Buldhana

Parameters	Details		
Soil type	Clayey		
Cropping details	Crop	Acres	Irrigation Method
	Cotton	1.5	Drip
	Wheat	1.5	Surface
	Sugarcane	1	Surface
	Onion	2	Sprinkler + Drip
	Potato	0.6	Sprinkler
Water source details	Type:	Open well	
	Depth:	78 ft.	
	Diameter:	18 ft.	
	Water column change:	10 to 50 ft.	
Pipeline details	Material:	PVC	
	Diameter:	2.5"	
	Length:	100 to 1300 ft.	
Irrigation infrastructure details	Sprinkler:	12 nozzles	
	Drip:	1 X 4 ft. spacing	
Existing pump set details	Capacity:	3 HP	

	Model:	CRI: CSM 45
	Head range:	28 to 19 m
	Flow rate range:	9 to 2 lps
	Overall efficiency:	-
Actual characteristics required for given configurations	Head range:	44 to 16 m
	Flow rate range:	2 to 6 lps
	Pump Capacity:	5 HP
Suggested model	CRI: CSM-3H	5 HP
	Head range:	45 to 17 m
	Flow rate range:	2 to 7 lps

This is the only case of undersized pump set. Taking into account the irrigation methods used for different crops and the distance between water source and field, a 4 HP capacity pump is required. But, since 4 HP pumps are not easily available in the market, a 5 HP capacity pump is suggested for this farmer. However, the farmer has installed a 3 HP pump set (CRI: CSM 45, BEE 5 star rated) and is satisfied with the performance.

This particular case needs a more detailed survey and on-field observations to understand the effect of land slope on pump performance and its ability to provide expected flow rates at respective heads.

Case V: Farmer 5, Junoni-Walgud, Osmanabad

Parameters	Details		
Soil type	Clayey loam		
Cropping details	Crop	Acres	Irrigation Method
	Soyabean	3	Surface
	Jowar	3	Surface
	Wheat	1	Surface
	Fodder	1.5	Surface
	Drumsticks	3	Drip (~1200 trees)
Water source details	Type:	Percolation tank	
Pipeline details	Material:	PVC	
	Diameter:	2.5"	
	Length:	1000 to 10000 ft.	
Irrigation infrastructure details	Drip:	10 X 10 ft. spacing	
Existing pump set details	Capacity:	5 HP	
	Model:	CRI: CSM 44 (ISI rated)	

	Head range:	43 to 11 m
	Flow rate range:	2 to 9 lps
	Overall efficiency:	40%
Actual characteristics required for given configurations	Head range:	42 to 16 m
	Flow rate range:	2 to 6 lps
	Pump Capacity:	5 HP

The field configurations are within the characteristics of the selected pump set and capacity is matching correctly. The required head value is on a higher side majorly due to frictional head as the water is getting transferred over a distance of 10,000 ft. (~3 km) from percolation tank to the field. This particular pump set is appropriately sized.

Case VI: Farmer 6, Junoni-Walgud, Osmanabad

Parameters	Details		
Soil type	Clayey loam		
Cropping details	Crop	Acres	Irrigation Method
	Soyabean	1.5	Surface
	Tur	0.5	Surface
	Onion	3	Surface
Water source details	Type:	Stream (till the Rabi season)	
Pipeline details	Material:	PVC	
	Diameter:	2.5"	
	Length:	200 to 300 ft.	
Existing pump set details	Capacity:	5 HP	
	Model:	CRI: CSM 2 LV (ISI rated)	
	Head range:	31 to 22 m	
	Flow rate range:	2 to 14 lps	
	Overall efficiency:	40%	
Actual characteristics required for given configurations	Head range:	10 to 8 m	
	Flow rate range:	6 to 7 lps	
	Pump Capacity:	3 HP	
Suggested model	Kirloskar: KOS 314+	3 HP	
	Head range:	14 to 6 m	
	Flow rate range:	9 to 14 lps	

This farmer has 3 acres of land and lifting water from a stream 300 ft. away from the farm. Hence, if a farmer's and vendor's criteria of IPS selection is used here, a 3 HP pump for 3

acres of land, surface irrigation with less than 2000 ft. distance is suitable. However, the existing pump set is a 5 HP pump, which is clearly oversized. The current operating head range (10 to 8m) is beyond the pump characteristics (31 to 22m). Hence, performance at these head values cannot be guaranteed.

However, the 7/12 extract and the PoCRA inspection teams report shows that, this particular farmer has selected a pump set to be used for an open well. Hence, more follow-up needs to be taken and an on-field survey is necessary to check if there is any elevation difference between the stream and the field. Also if the pump is being used for water transfers from stream to well.

2.6.2. Summary of findings

- 29% pumps (9 out of 31 samples) were found to be oversized. Installing correctly sized pump sets for these configurations can lead to a reduction of at least 13% load (~2 kW per pump).
- Only 2 out of 31 sample pumps are found to be selected for the correct operating head range (case II and V) that also matches the pump capacity.
- From the feedback from the farmers, the reasons for selecting higher capacity pumps as told by farmers are:
 - Drip or sprinkler system needs very high pressure heads and farmers want to operate more number of nozzles at one time
 - Due to unreliable supply of electricity: farmers want to ensure high flow rates whenever there is power supply
 - If a farmer needs 3 HP pump, but can get a 5 HP pump from a subsidy with similar investments of about (Rs. 4000 to 6000/-), a higher capacity pump is selected.
 - Because of MSEDCL's tariff structure, a minimum load of 5 HP is considered. (0 to 5 HP, 5 to 7.5 HP, 7.5 to 10 HP).
 - 4 HP pump set is found to be suitable for 4 out of 31 farmers. However, pump sets having capacities other than 3, 5, 7.5 HP are not available in the market. The reasons for the same need to be analysed further.
- One case of undersized pump (case study IV) was found in Buldhana. For the farmer the existing 3 HP pump seems to be working fine, but as per the field configurations,

selecting a higher capacity pump is recommended. One of the reasons could be the existing pump set is a BEE 5 star rated pump. This particular case requires the on-field survey, as the exact land slope and terrain details are unknown.

- Operating head range does not match with the recommended characteristics even if the pump set HP is appropriate. This results in lowered operational efficiency, and ~10-15% higher energy consumption than required.
 - Pumps that are used for water transfers over long distances are found to be appropriately selected and fitting the given head range. This is mostly due to high frictional head value.
 - The savings for specific farmers have been calculated in the case studies. But considering the broader picture of current practices in the state, then a considerable amount of energy and costs associated with it can be saved at all levels (farmer, MSSEDCL and Government).
 - Since pump capacities can be reduced for 25% of the pumps, more number of farmers can be connected to the grid for a certain cost. Cost of pumps, whether paid for by government schemes for farmers, goes down.
-
- Pump set standards can play a crucial role in improving the overall pumping systems efficiency. The new IS 14220:2018 open well submersible pump set standard states the parameters to be marked by the manufacturer on the pump set. It includes details about rated voltage, duty point, overall efficiency, operating head range (BIS, 2018). However, in addition to this the labelling needs to include variation of flow rate with head, variation with voltage, operating voltage range instead of stating only the best operating point. This is because, in a practical scenario, hardly any pump will operate at the given best operating point. This will also help pump vendors and the farmers to select a pump which is suitable for the actual on-field configuration.
 - There is a wide head range at which the pump operates practically, considering the variation in irrigation methods and the season of irrigation. A feedback mechanism between the farmers/ vendors and the pump manufacturers should be there to optimise this structure of practical requirements.
 - Along with defining proper standards and feedback to manufacturers and the government, increased awareness among farmers will make it easy to improve the

systems performance as the problem of improper selection of pump sets will be taken care of up to some extent.

- There is no generalized methodology to improve the effectiveness of the overall system. Different options are available and have been studied in the past for improving the effectiveness from selection of appropriate pump set to replacing the complete system with more efficient.
- Agriculture demand side management (AgDSM) is one such project that was implemented on a pilot scale across the country. AgDSM involved two components, one is the rectification and replacement of pump accessories (pipelines, valves, etc.) and the other is replacement of the entire pump set. Studies from both approaches reported about 19 to 34% energy savings and increased energy consumption by 11 to 12% due to increased pump discharge (Khobragade, 2018).
- But, a major problem faced in these AgDSM projects is the problems with power supply quality and access. The quality of power supply is a critical issue not only affects the pump selection practices by farmers but also results in inefficiency of IPS. Hence, energy infrastructure development must be conjointly done.

2.6.3. Way forward

A comprehensive and synchronized approach is needed to highlight and deal with the challenges at each level. We need to discuss with various stakeholders in the IPS ecosystem to understand their response to our findings from the field and suggested guidelines.

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3. Irrigation infrastructure: Water transfer structures

Analysis of water transfer structures is crucial from both energy and water perspectives as it is further increasing the competition for water and energy resources. Factors documented and analysed in this study are the reasons for transferring water between two sources, seasonal as well as regional variation in type and frequency of transfers, crops for which transfers are being done specifically, and the energy associated with it.

We define water transfers as, when water is not pumped directly from the water source into the field or it is pumped over the longer distances. There are 3 main types of water transfers as indicated below:

- Type I: Pumped out of open well or bore well or canal in Kharif (or during a rotation), and transferred to farm ponds for use in Rabi.
- Type II: Pumped from a borewell into a shallower source such as open well or farm pond and thereafter pumped into the field, so as to provide a high enough discharge required for furrow irrigation or to use more nozzles during sprinkler irrigation.
- Type III: Pumping long distances (in km) from a river or dam or irrigation tank into the field.

For Type II transfers, a detailed analysis has been done which involves identifying the reasons for looking for more than 1 water source, additional investments in it, their benefits to farmers, and estimates of energy usage in each of these types of water transfers.

At the end of the section, for Type III water transfers an estimation of energy consumption for individual lift irrigation method and potential energy savings based on the findings from previous studies on cooperative lift irrigation schemes are included.

3.1. Washim village cluster

There are two major types of water transfers seen in Washim village cluster, borewell to open well and minor irrigation tank to the fields. The data has been collected from 3 villages, Umbarda Bazar, Yevta and Manbha as these are connected by the same Ag feeder network.

Type II: Borewell to open well water transfers

General points:

- Open well is used as a storage tank to store the water pumped from borewell. The average borewell depth is 200 to 500 ft.
- 70 to 80% of the farmers having both borewell and open well, transfer water from borewell to open well. This water transfer is due to the low flow rate when water is pumped from the borewell. If borewell is used directly, it takes more time to irrigate with a surface irrigation method and when connected to drip or sprinkler infrastructure the desired number nozzles and emitters cannot be used.
- In general large farmers are also likely to have both borewell and open well, and hence water transfer phenomenon is common with them.
- During the start of the Rabi season, very few cases have been observed where borewell is directly used as a water source for the first 2 watering. After that, water gets transferred to the open wells due to low flow rate as the water table goes down further.

Table 7: Disparity in registered and actual on-ground data

Village	Number of borewells		Number of open wells	
	registered in Talathi office	Feedback from farmers	registered in Talathi office	Feedback from farmers
Umbarda	24	30 to 40	190	> 200
Yevta	40	>100	-	-
Manbha	56	>100	412	400 to 450

*Depth of bore wells and open wells was found to be varying from 200 to 500 ft. and 25 to 60 ft. respectively.

Reasons for looking for more than 1 water source and water transfers:

- The number of borewells as a source of irrigation are increasing as it saves time as well as money. Every year an attempt is made by 4 to 5 farmers to dig a borewell. In the last 5 years, the number of borewells have increased 3 fold in these villages.
- There are two important reasons for this, one is to satisfy the crop water requirements of already sown crops (Wheat, Tur, Turmeric, Sugarcane, etc.). The other reason is to cultivate more crops and improve the returns. The water transfer starts from the month of December and lasts until April-May. But, the water availability decreases (subsequently hours of pumping gets reduced) as we go from December to May.
- Another reason is low yield of open wells and the water is not sufficient. Some of the open wells were reported to get dried up even in the month of January. (not able to provide last 1 to 2 watering to Wheat)
- For some percentage of farmers, the piping to the field, or drip /sprinkler infrastructure, is connected from a main water source (like open well), hence water from other source(s) is first transferred into the main source and then to the fields.
- Most of the farmers need water for only 3 to 4 months (in Rabi season) to irrigate Cotton, Tur, and Wheat. And if the open well yield is low, they are not able to satisfy the crop water needs, hence they attempt to dig a well.

Energy requirement:

- Minimum 5 HP capacity pump has been installed by most farmers.
- Borewell is in operation from November till May for the 4 to 8 hours of electricity availability. Number of hours of operation vary depending on the crop and the season. Farmers generally find it convenient for them to transfer water from borewell to open well during the night time electricity supply.
- According to front-line workers in MSEDCL, water transfers from one source to another might be 30-40 percent of the load in Karanja sub-division (season not specified).
- The estimation of energy consumed in water transfers for major crops in the Washim cluster is given in Table 8. Energy required for each crop is calculated based on the average irrigation depth, hours of irrigation in different months, volume of water transferred, total number of irrigations, assuming a 5 HP capacity pump set and 90% distribution losses.

Table 8: Estimation of energy consumption throughout the season for major crops in Washim village cluster

Crops	Energy Consumption for direct irrigation (without transfer)		Energy Consumption with water transfers	
	kWh/ acre	kWh/ unit volume of water	kWh/ acre	kWh/ unit volume of water
Soybean (Kharif)	61	0.14	-	-
Cotton (Long Kharif)	184	0.2	278	0.48
Tur (Long Kharif)	92	0.14	139	0.41
Wheat (Rabi)	367	0.22	1039	0.65
Gram (Rabi)	83	0.24	157	-
Orange (year round)	673	0.29	1790	1.11
Papaya (year round)	300	0.29	831	1.11

Table 8 shows the large variation in kWh/ acre of crop sowed. The energy consumed per acre (kWh/ acre) is found out to be very low in case of Soybean (mostly considered as a rain-fed crop and irrigation is given only in the dry spells if needed), but for Cotton and Tur (long Kharif) around 40% increased consumption can be seen (few water transfers are involved for irrigations done in October to December). From Rabi crops such as Wheat and Gram, the ratio of energy in water transfers to energy for direct irrigation per acre is about 60%.

In case of kWh/ volume of water used for irrigation, the energy consumed is almost 3 times for farmers doing water transfers. And in the case of Orange, the energy consumed per unit volume transferred and used is likely to be higher than 1.1 kWh. The energy used for water transfers are estimated for horticulture crops and shown in Figure 6.

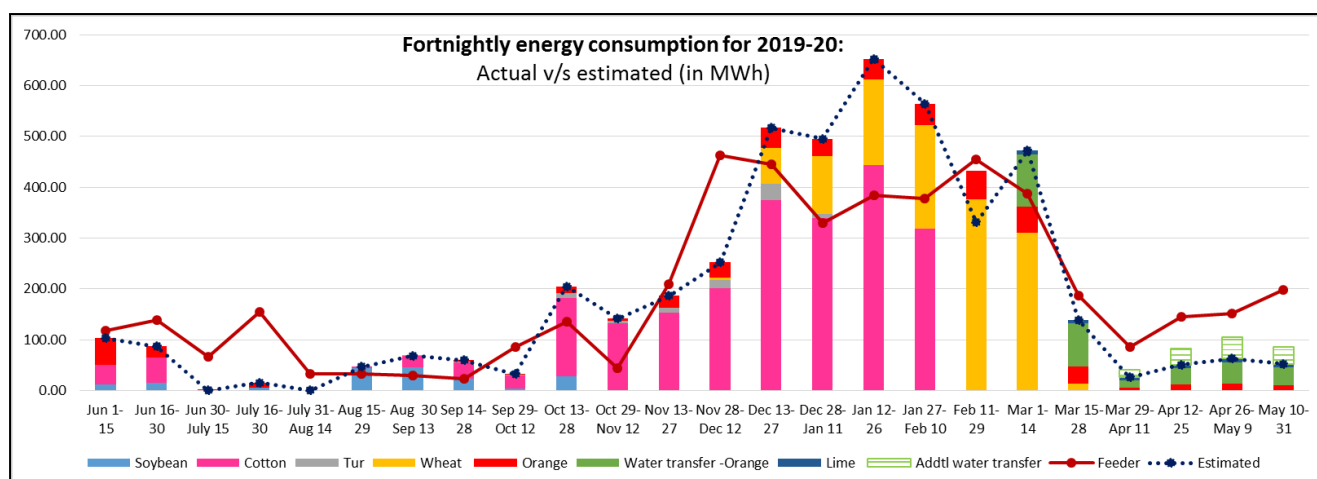


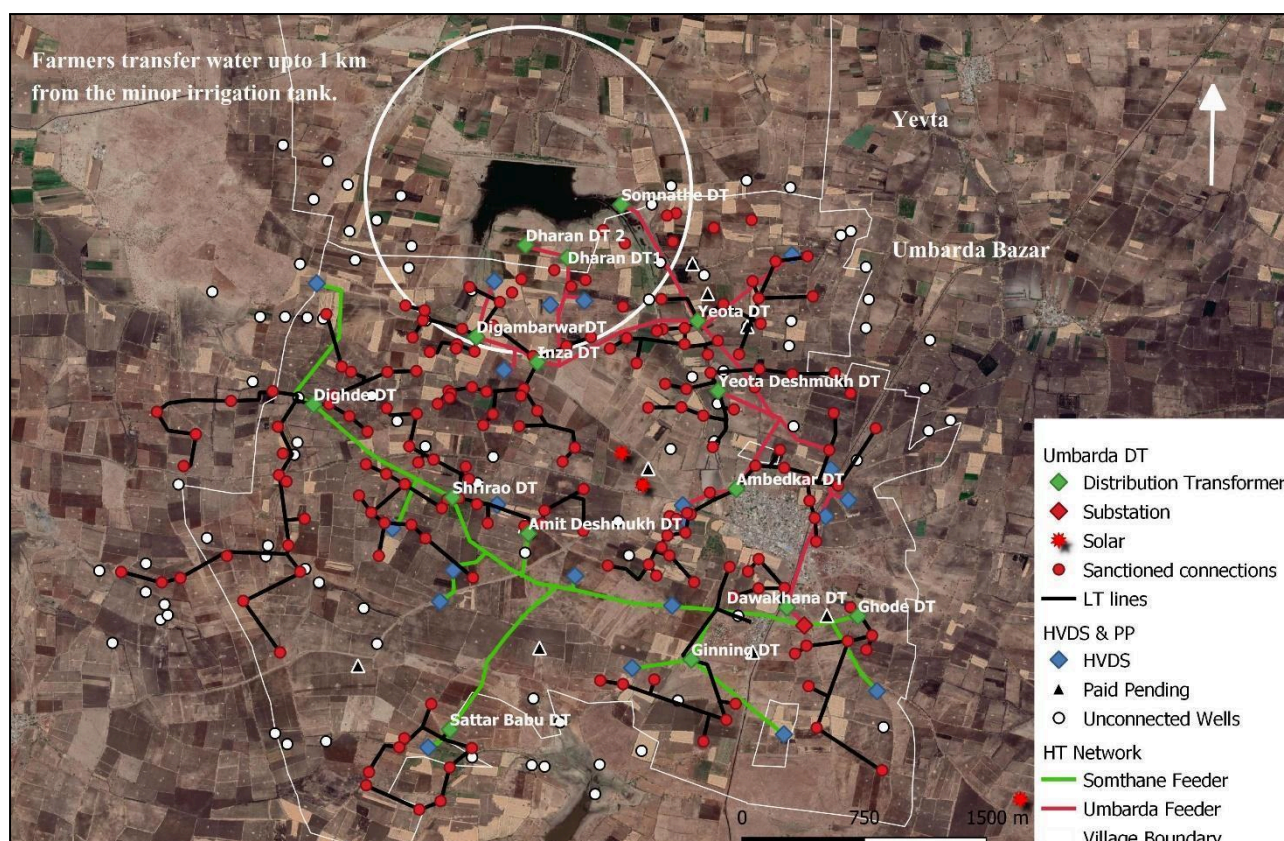
Figure 6: Estimated v/s actual energy consumption (including water transfers)

Investments:

- It takes 4 to 5 months to dig an open well. Cost of digging the open well varies from Rs. 3 to 6 lakhs depending on the depth of the well, type of soil, etc. This is more time consuming, requires high investment and planning.
- On the other hand, only a day is needed to dig a borewell and costs at Rs. 90/- per foot. And additional cost of pipes and electrical cables (Rs. 30,000 to 50,000/-) and (Total cost less than Rs. 1 to 2 lakhs).
- However, it is highly unreliable that the water is available where the borewell is dug. One of the farmers reported that 4 out of 5 attempts of borewell were failure events with each attempt costing between Rs. 20,000 to 30,000 /-.
- But, as mentioned above, most farmers need water only for 3 to 4 months (in Rabi season). If the bore fails they can dig up another one, but constructing an open well does not provide this flexibility. This is one of the reasons for the rapid increase of unaccounted load in the last 10 years. Even if the borewell is not successful at first attempt, farmers try multiple times at different locations or in the same bore to greater depths.

Type III: Minor irrigation tank to field

- Yevta minor irrigation tank has a water storage capacity of 2.39 million metric cube and irrigation capacity of 377 hectares. (as per completion report of irrigation department)
- About 70 to 80 farmers from Umbarda (43) and Yevta (33) (total 76 in year 2019-20) are currently lifting water from the tank.
- Major crop irrigated by lifting the water from tank in year 2019-20 is Wheat with an area of 32.9 hectares. (~9% of reported irrigation capacity) (as per Rabi crop report 2019-20, irrigation department)
- Farmers transfer water over a distance of a few meters to more than a kilometre.
- Water is available till the month of February-March. (3 to 4 months of water usage)
- Farmers have to pay water tax for lifting water from the tank. The fees are charged based on the per hectare land to be irrigated. But, there is no limit on water to be lifted from the tank. The irrigation department is not able to monitor the water usage by each farmer as it is a complex, time consuming process.
- While giving permission to lift water from a tank, the irrigation department does not keep a check on farmers if they have legal electricity connections or not. Also, there are no guidelines to limit the capacity of pump sets used for lifting water and the distance to which the water gets transferred.
- Hence, access to water is renewed every year, but the energy availability and access is taken for granted.



The circle shows the area under Rabi crops that gets irrigated by minor irrigation tank water in Umbarda and Yevta villages.

Analysis of energy usage for Type III water transfers in Umbarda and Yevta

The analysis of energy consumption is done only for the Wheat as only Wheat is listed in the records of irrigation department. Number of farmers who lifted water in the year 2019-20 for Rabi season and the total area irrigated is taken from the annual reports of irrigation department. For the estimation of energy consumption, two approaches have been used. The approach is based on the understandings from on-field measurements of irrigation depth, water used (in m³) and flow rates.

Table 9: Estimation of energy consumption for Type III water transfers in Washim cluster

Crop	Wheat
Area irrigated (hectares)	32.9
Irrigation depth (mm)	50
Water usage (TCM)	132
Flow rate (lps)	5

Overall efficiency (%)	20
Hours of irrigation	762
Total number of irrigations	8
Energy consumption (MWh)	18
Energy consumption (kWh/ m3)	0.20

Average distance of water transfers is considered to be 1500 ft. (500m)

3.2. Yavatmal village cluster

- Yehala and Kondhai villages have 3 different water sources viz. dam, open well and borewell. Number of borewells are very less in both villages.
- About 90% of the farmers in Yehala (~200) and 10-15 farmers in Kondhai (total 210 in year 2019-20) are currently lifting water from the tank lift water from the Adhar Pus Dam. Average length of water transfer is 2000 ft. but there are few farmers having pipeline length up to 6000 ft.
- Major crops irrigated by lifting the water from the dam in year 2019-20 are given in Table 10. (as per Rabi crop report 2019-20, irrigation department). Farmers have started cultivating crops such as Banana, Sugarcane and Turmeric, but the percentage is very low (only 3 to 5 hectares area).
- Farmers have to pay water tax for lifting water from the dam. It is charged based on the per hectare land to be irrigated for different crops. But, there is no limit on water to be lifted from the tank. The irrigation department is not able to monitor the water usage by each farmer as it is a complex, time consuming process.

Table 10: Details about area and energy consumption for major crops in Yehala and Kondhai

Crops	Area Irrigated (Ha)	Water usage (TCM)	Energy Consumption (MWh)
Cotton	20	20	4
Wheat	71	248	38.1
Gram	130	104	15.7
Groundnut	36	130	13
Others (Turmeric, Horticulture, etc.)	12	79	18.4
Total	257	502	89.2

Average distance of water transfers is considered to be 2500 ft. (800m)

We looked at the data for 7 cooperative lift irrigation schemes for Sugarcane in Kolhapur and Sangli districts. For 5 of the cooperatives, average energy consumption was 2 kWh/ ha-mm. For the remaining 2 cooperatives, the consumption was 5 kWh/ ha-mm and 6 kWh/ ha-mm respectively. This was majorly due to the small pipe diameter used for transferring water which increases the frictional head.

Contrary to this, for individual lift irrigation, we observed that on average the energy consumption was 3 kWh/ha-mm for the same terrain.

This indicates that at least 30% energy can be saved by implementing a cooperative irrigation scheme. In the case of Washim and Yavatmal village clusters also (individual lift irrigation), very high energy consumption has been observed. Hence, it will be worthwhile to do a feasibility analysis of implementing a cooperative scheme at these locations.

3.3. Osmanabad village cluster

- The borewell depth in a Junoni - Walgud village ranges between 200 to 700 ft. and the average depth of open wells is 40 ft.
- About 60-70% farmers have at least 1 borewell in addition to open well irrespective of land holding size.
- It is observed that borewell water is mainly used to ensure Rabi crops (as open wells are on the verge of getting dry in November and have lesser yields) and to support year round crops.
- Farmers not having open well, have preferred borewell as the capital cost for borewell is 4 times less than that of digging open well.
- The borewell pumping hours in Rabi season varies between 4 to 8 hours as per requirement of crops sown (average twice a week). This results in energy consumption of average 800 kWh per farmer.
- The borewell pumping hours in a summer limits to 2 to 3 hours as availability of water decreases. This average energy consumption is 230 kWh per farmer.
- One of the farmers took help from Groundwater Survey & Development Agency, GoM before BW construction. Their investigation showed that water holding capacity of stratum in a Junoni village is less, hence the stratum cannot store water effectively.

4. Estimates of water and energy off-season consumption [H]

The crop water requirements for major crops in the Washim cluster are calculated using the water balance model. Also, FGDs and in depth interviews were carried out with farmers to understand their irrigation practices. Based on this, energy consumption for each crop was estimated. This exercise is important because the estimation will help us understand the on-ground scenario and identify the reasons for differences between actual versus estimated consumption. After classifying the focus areas, a more refined model can then be developed, validated and used to gauge the extent of energy security and access in other parts also.

4.1. Executive Summary

- The estimated energy consumption curve follows the similar trend as that of actual energy consumption on both Umbarda and Somthane feeders, except for July and December to January period. Some of the reasons for the difference between actual and estimated consumption could be assumption that Wheat sowing was done in December and interrupted supply in peak Rabi months.
- Estimation of energy usage for horticulture crops including water transfers is 972.3 MWh, for Wheat it is 1261.5 MWh and Cotton the energy usage is 1606.3 MWh.
- 40% of the farmers do water transfers from borewell to open well for crops such as Wheat, Orange and Papaya. Total energy usage for these transfers is estimated to be 435 MWh.
- The area under horticulture in Washim cluster is increasing in the year 2020-21, and hence the demand for energy is also increasing.

4.2. Details about methodology used for estimation:

- Feeder energy consumption data for Umbarda and Somthana Ag feeder in Karanja Lad, Washim for year 2019-20 (June 2019 to June 2020).
- Villages: Total 9 villages, Umbarda Bazar., Yevta, Manbha, Pilkheda, Sukali, Dudhora, Pimpri Warghat, Somthana, and Wadgaon
- Soil type: 90% Clayey, 5% Clay loam and 3% and Gravelly clay loam
- Water sources:
 - Open wells: usage throughout the season, storage tank during water transfers

- Borewells (November to May): 30 to 40% of the farmers using borewell for water transfer to open well. Crops: Cotton, Tur, Wheat, Gram, Orange, Papaya, Lemon
- Yevta irrigation tank (November to till Feb/ March): Water transfer distance varies from few meters to more than 1 km. Crops: Cotton, Tur, Wheat, Gram
- Cropped area information used as received from Krishi office indicated in Figure 7. Area under horticulture crops can be seen increased in year 2020-21. Other category includes Vegetables, Sugarcane, Turmeric, etc.
- Crop water requirement calculations are done based on a point water balance model that computes the generation of stocks and flow arising from the water cycle.
- Water transfers are taken into account from the month of December and continue till May in case of Cotton, Tur, Wheat (Figure 8) and Orange, Papaya (Figure 9). No water transfers are considered for Lemon as there are very few farmers who need to do that (as per feedback from a sample of farmers growing Lemon, Yevta).
- Pump efficiency varies from 5% in monsoon (for irrigations during dry spell, August-September) to 30% in January and then further reducing.
- The estimation assumes that 50% cropped area is irrigated by borewells from mid-December which is a common practice in the region. This is based on the FGDs and primary surveys conducted in the month of February 2020. However, 2019 and 2020 have been good rainfall years, bore-well usage is likely to have reduced and/ or delayed. This will be verified in the Rabi season field visit and the estimations will be adjusted according to that.

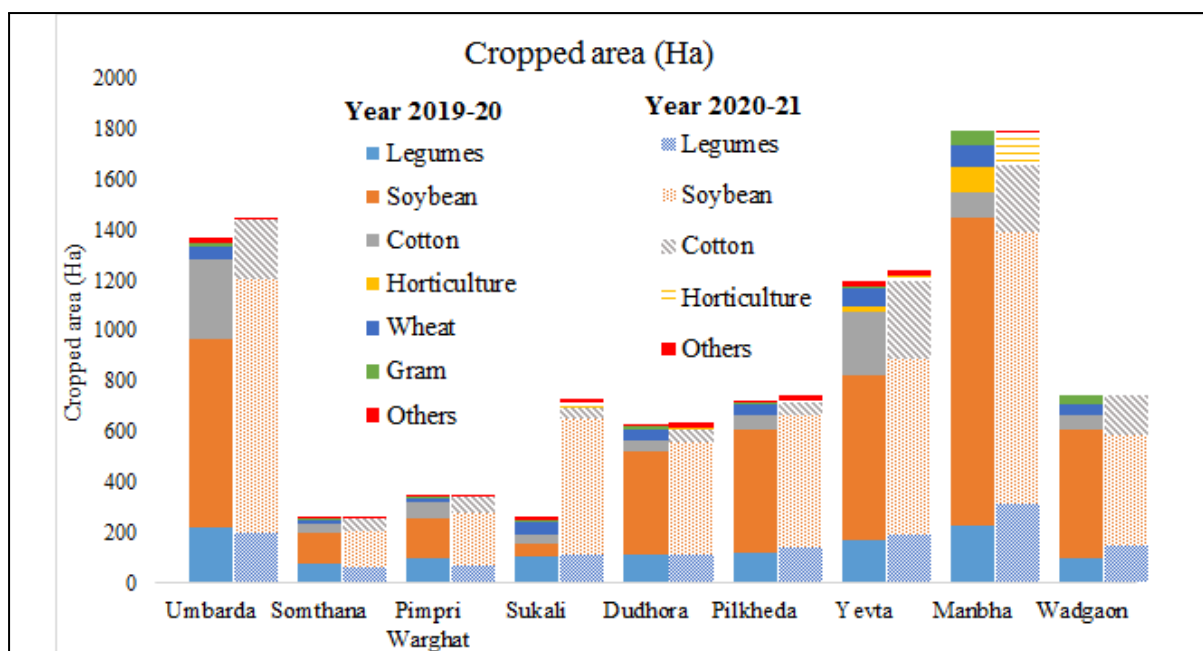


Figure 7: Cropped area for year 2019-20 (Kharif and Rabi) and for year 2020-21 (Kharif)

4.3. Fortnightly energy consumption for year 2019-20

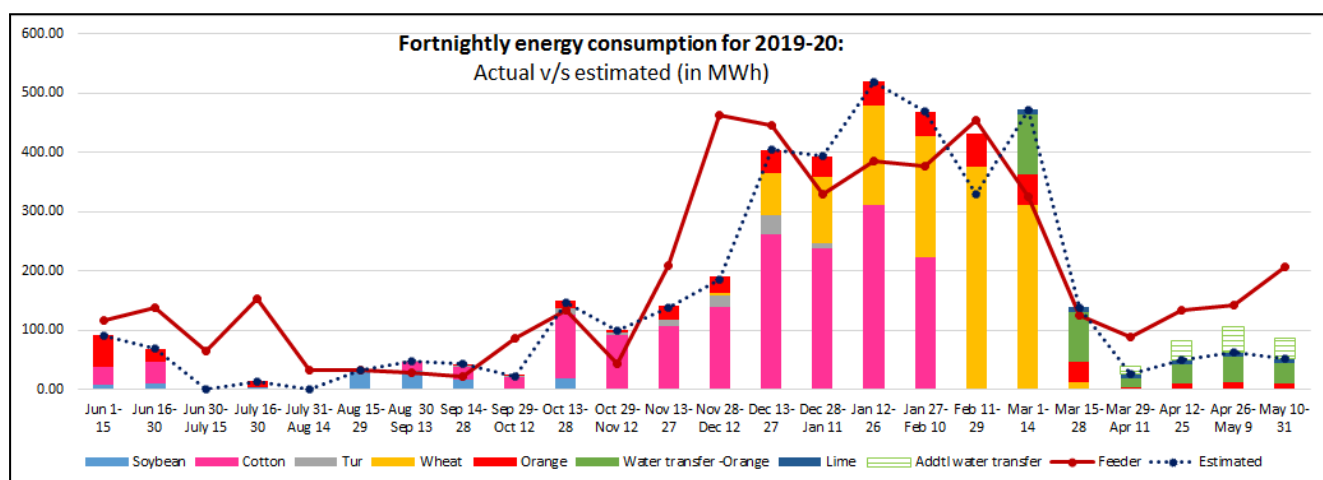


Figure 8: Crop wise estimated v/s actual energy consumption

Major demand for irrigation and energy consumption is in Rabi season for Cotton, Tur, Wheat, and other Horticulture crops. Peak energy demand was calculated from December to February considering the crop water requirement and water transfers, but the actual feeder energy consumption was observed to be less. One of the reasons could be that actual water usage differs for farmers based on the water availability in their wells and climatic conditions. Also, there are farmers who do not irrigate Cotton and Soybean (rainfed). Hence,

only 70% of the area under Cotton and Soybean is taken into consideration. This particular assumption will be verified with the Krishi office and farmers in next field visit.

On the other hand, for summer months the actual feeder energy consumption is seen much higher than estimated. The exact reason is not known at present, but one of the reasons could be more than 1 water transfer before actual irrigation. For example, few of the farmers have more than 1 borewell and transfer water from both borewells to open well. In that the energy consumption for water transferred will be twice of what has been considered now. The Green colour boxes represent this additional energy involved in water transfers.

Interrupted supply in peak Rabi months due to DT failures and breakdowns could also be the reason for the difference between actual and estimated energy consumption especially for December to January. A small farmer might irrigate the same parcel of land again if he was not able to irrigate in the first go, however for the medium to large farmers the entire area might not get irrigated, in spite of using multiple pumps. The difference was also observed at the start of December (Nov 28 to Dec 12), this is because we have assumed that the Wheat sowing will start in December, but some farmers also start at the end of November or irrigate their fields before sowing.

Other than this period, the estimated energy consumption curve follows the similar trend as that of actual energy consumption on both Umbarda and Somthane feeders.

4.4. Summer or Offseason consumption

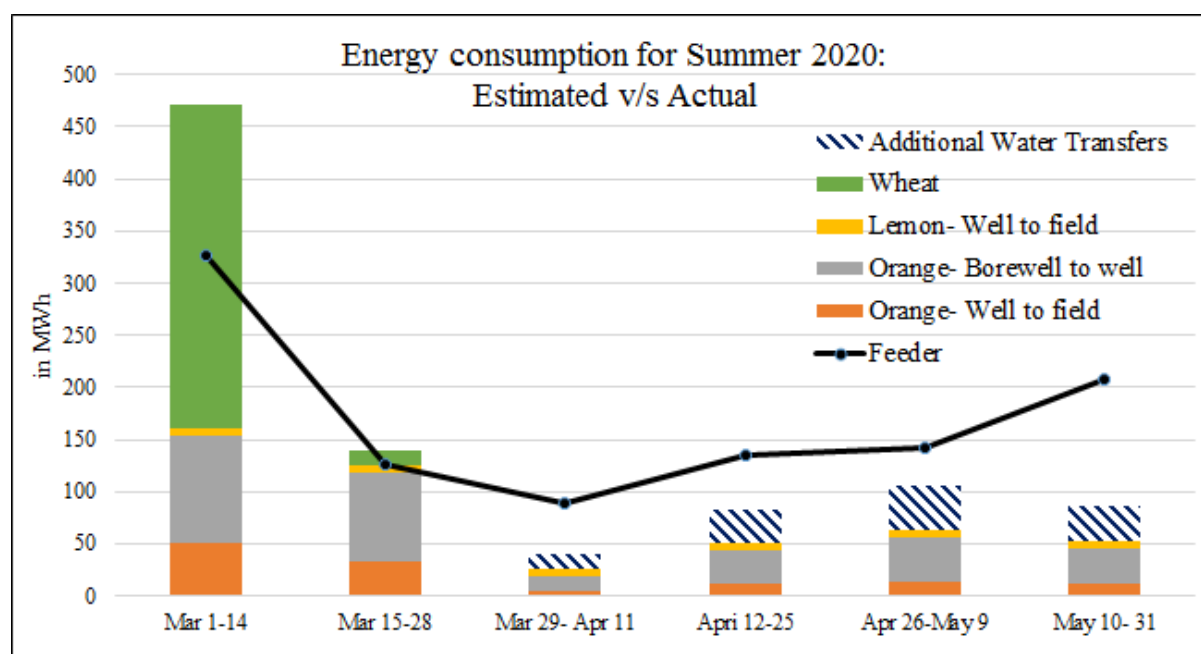


Figure 9: Energy consumption in summer months (2020)

The crop water requirements are calculated based on the water balance model and rainfall data of year 2019. In summer 2020 (Figure 9), a significant rainfall (46 mm) was recorded during the months of February and March. This resulted in a decrease in crop water requirement, reducing the energy demand in that period.

Orange and Lemon are the only crops in summer, the energy involved in actual irrigation can be seen as more or less the same each fortnight. Major share of the energy is going into water transfers.

There is still a gap between estimated energy consumption versus actual feeder consumption. One of the reasons could be that, in the summer season, we have assumed that water transfer is being done from 1 borewell only (1 borewell to 1 open well). However, as the farmer surveys are evolving in every visit and it seems there could be more than 1 borewell that is used by farmers for water transfers. (1 of the farmers has 3 borewells and 1 open well). To satisfy the crop water requirement in summer months, the energy consumption per unit of water pumped will increase as the water table also drops.

Farmers are not upfront about telling the number of borewells they use making it difficult to estimate the water and energy usage. Also there is great disparity between the number of registered borewells in Talathi office and actual number of borewells (Table 7).

To show the consumption that could be possible with these extra borewells, additional water transfers are calculated and added in summer months (blue hatch) to show if there were 1 more additional borewell for all farmers, how much consumption it could be.

The aggregation in March first and second week is seen considering the last irrigation for Wheat for some of the farmers. This particular phenomenon will be verified and the estimation will be refined after the field visit.

To comment on the actual water usage by farmers' one round of measurements was done in Kharif 2020. Following that, investigations will be done in Rabi and Summer 2020 season as well as the data from installed energy meters will give us a clear picture of water and energy usage patterns.

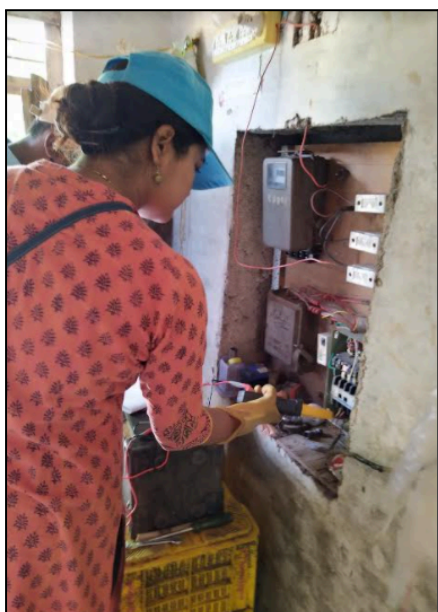
● Appendix I

Standard protocol for recording measurements:

- **General survey:** Crop, cropped area, crop spacing, method of irrigation, soil type, water source dimensions, pipeline details
- **Measurements:** Voltage, current, power, power factor, head, flow rate, length of pipelines

Sr · No.	Pum p set type	Ma ke	Rated Details				Measured details				Overall Efficienc y (%)
			Pow er	He ad	Flo w rate	Overall Efficienc y	Pow er	Stat ic Hea d	Flo w rate	Velocit y	
1											
2											
3											
4											
5											

*Duty point details of some of the open well submersible pump sets were not available.



● Appendix II

Table 11: Detailed calculations for input power required for given configuration

Crop	Area (acre)	Irrigation method	Flow rate (lps)	Head (m)	Irrigation depth (mm)	Hydraulic Power (kW)	Overall Effie. (%)	Input power reqd. (kW)
Cotton	2	Furrow	6	15.2	50	0.8	0.35	2.4
Cotton	2	Sprinkler	3	35.0	6	0.9	0.35	2.4
Tur	4	Furrow	6	19.6	50	1.1	0.35	3.1
Tur	1.5	Drip	2	31.5	20	0.7	0.35	2.0
Wheat	3	Furrow	6	15.4	50	0.8	0.35	2.4
Gram	2	Sprinkler	3	38.0	12	0.9	0.35	2.7
Guava	1.5	Drip	2	35.0	20	0.8	0.35	2.2

Table 12: Calculations for energy usage and savings for configurations in Table 11

		Existing pump set: 5 HP	Existing pump set: 5 HP
Time reqd. for 1 irrigation (hr)	No. of irrigations	Energy usage (kWh/Ac)*	Energy usage (kWh/Ac)*
20.2	5	253	152
5.4	2	27	16
20.2	3	76	46
14.9	3	149	89
30.4	8	405	243
10.8	2	54	32
14.9	90	4459	2676
	Total Units	5422	3253
	Energy savings (kWh/ year)		2169
	Cost savings (Rs./year)		8,676.00

*HP ~ kW

Table 13: Details of instruments used during field measurements

Power Analyser	Ultrasonic Flowmeter
- Model: DT369 - Make: MEXTECH - AC Current 4A, 40A, 400A/600A - AC/DC Voltage 600V (Auto Ranges) - Resistance 40MΩ (Auto Ranges) - Capacitance 100uF (Auto Ranges).	- Portable Hand held Ultrasonic Flow Meter - Make: Base, Model: BHUFM1000 - Flow Rate (m3/hr or LPM or LPH) - Accuracy: ± 1 % of measured value - Operating temperature: -40 to 150 °C - Transducer size: M type (50 – 700 mm) (2” – 28”)

Table 14: Description of energy and water meters installed

Energy meters	Water meters
- 3-phase whole current LT energy meters for agricultural pumpsets - Monitor at a minimum integration period of 15 minutes - Load data for 60 days - Energy consumption in kWh, kVAh; V, I, parameters on all three phases - Model : Secure make SARAL 300, 10-100A, Class 1.0, BLE - Custom made with 1 month lead time - remote data loggers (7 meters in each village, total 42)	- Accuflow make, Woltman Type - Removable Mechanism Water Flow Meter - Regular water meter readings to be done manually (6 meters in each village, total 36) - Cumulative volume recording

● References

- BIS. (2018). *Openwell Submersible Pumpset Specification (First Revision) (IS 14220 : 2018)*. Bureau of Indian Standards.
- Khobragade, S. P. (2018, June). *Investigating Agriculture Demand Side Management through ESCO – (A case study of Solapur Pump replacement Project)*. Retrieved from Centre for Technology Alternatives for Rural Areas, CTARA: (<http://www.ctara.iitb.ac.in/en/mtech-project/>) (Last accessed on:22/07/2020)
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